



FEDERAL SERVICE
FOR INTELLECTUAL PROPERTY

(12) **ABSTRACT OF INVENTION**(21)(22) Application: **2016108260, 09.03.2016**(24) Effective date for property rights:
09.03.2016Registration date:
30.05.2017

Priority:

(22) Date of filing: **09.03.2016**(45) Date of publication: **30.05.2017** Bull. № 16

Mail address:

**620002, g. Ekaterinburg, ul. Mira, 19, UrFU, Tsentr
intellektualnoj sobstvennosti, Marks T.V.**

(72) Inventor(s):

**Fedosov Egor Aleksandrovich (RU),
Goldshtejn Sergej Lyudvigovich (RU),
Diomidov Ilya Andreevich (RU)**

(73) Proprietor(s):

**federalnoe gosudarstvennoe avtonomnoe
obrazovatelnoe uchrezhdenie vysshego
obrazovaniya "Uralskij federalnyj universitet
imeni pervogo Prezidenta Rossii B.N. Eltsina"
(RU)**(54) **METHOD FOR DETERMINING THICKNESS OF FLAT LAYER**

(57) Abstract:

FIELD: measuring equipment.

SUBSTANCE: essence of the method consists in the thermal impact on the surface flat layer of the living tissue-type of Hemangioma to determine its thickness. Herewith the unilateral pulse effect on Hemangioma by the thermal flow is achieved through creating a negative heat flow with a set of ice plates, which are consistently placed on Hemangioma, moisture is cleaned

after complete melting of each plate, the registration of the temperature change on the Hemangioma surface is carried out step by step along the lected direction, the desired thickness is calculated by the formula.

EFFECT: objective identification of the surface layer thickness with an accuracy of 5-10 percent on the living tissue-type of Hemangioma.

1 dwg

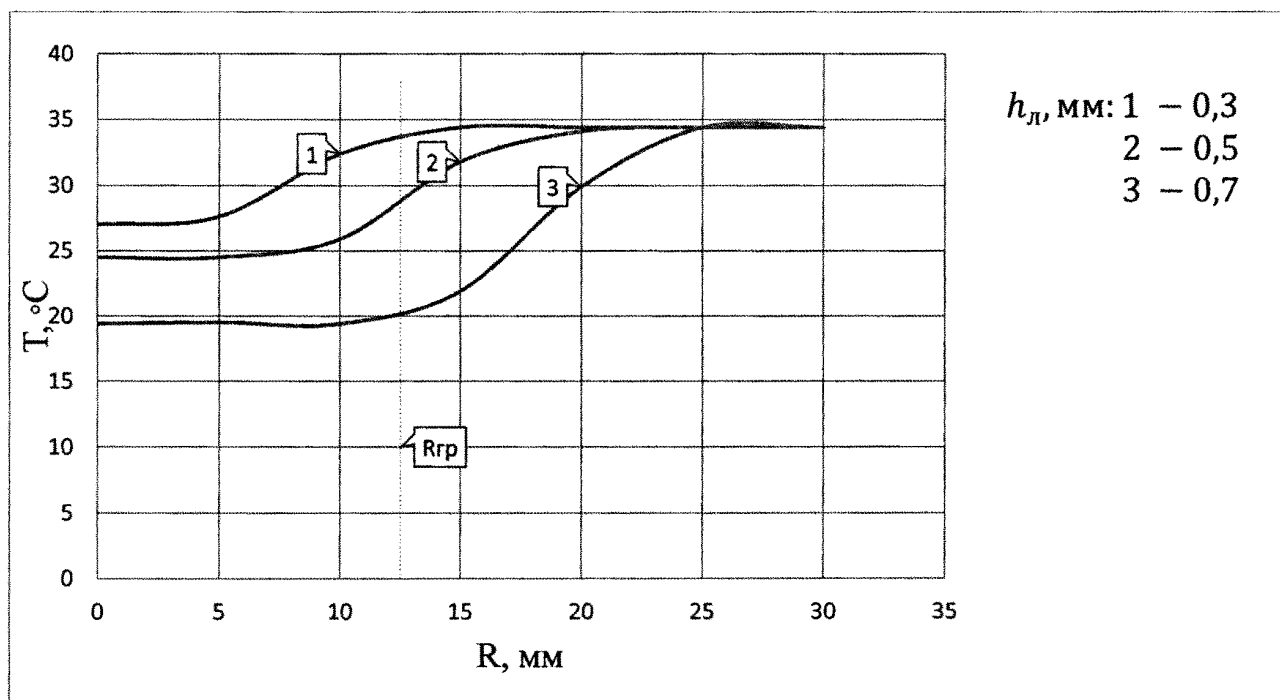


Fig. 1

The invention relates to measuring technology, namely to methods for determining the thickness of flat layers, primarily living tissues, using their thermophysical properties.

Known analogues include:

- 1) I. V. Sergeeva. Use of the interaction of a thermal pulse with a homogeneous plate for nondestructive testing [Text] / Defectoscopy. - 1984. - No. 4. - pp. 37-42;
- 2) Method for determining the thickness of a material layer, Russian Federation Patent No. 2023237 / V. D. Korneev [Electronic resource]. Available at: <http://www.freepatent.ru/patents/2023237>;
- 3) Method for determining the thickness of a flat layer, USSR Inventor's Certificate No. 1557454 / V. N. Tretyakov [Electronic resource]. Available at: <http://www.findpatent.ru/patent/155/1557454.html>.

At present, the thickness of living tissue is generally determined using scanning devices, which entails various costs, including labor costs. The body is also exposed to radiation during such examinations.

The essence of the methods presented in all the analogues consists in applying thermal action to a metallic flat layer in order to determine its thickness.

Analogue 3, adopted as the prototype, is the closest to the proposed solution. This method is intended for determining the thickness of flat layers, primarily metallic ones, using their thermophysical properties, and comprises three operations.

The first operation is the application of a pulsed heat flow to the layer. The second operation is recording parameters by registering a temperature change on one surface; such temperature registration is performed either during the thermal pulse or after its completion, and the rate of surface heating or cooling is determined. The third operation is determining layer thickness from a stable region of an auxiliary function with calculation according to a formula.

The known prototype has the disadvantage that it does not account for the specific features of operation on living tissues, particularly hemangioma-type neoplasms.

The objective of the invention is to determine the thickness of a superficial hemangioma-type tumor by a nondestructive method.

To achieve this objective, the following method is proposed. The proposed method for determining the thickness of a flat layer comprises applying a unilateral pulsed heat flow to the layer, constant throughout the pulse and uniformly distributed over the layer surface, registering a temperature change on the layer surface, and determining its thickness with allowance for the absorbed heat flow and the heat capacity of the layer material, and is characterized in that a hemangioma-type living-tissue layer is used as the flat layer; before the unilateral pulsed heat-flow action, copying material is placed on the hemangioma, its outline is recorded thereon, and a copy is produced; the center of mass of the copy is found, and the direction and distance from the center to the most distant point of the outline are determined; a temperature sensor is fixed above the hemangioma; the unilateral pulsed heat-flow action on the hemangioma is performed by creating a negative heat flow using a set of ice plates sequentially placed on the hemangioma; after each plate has completely melted, moisture is removed; the temperature change on the hemangioma surface is registered stepwise along the selected direction, and the required thickness is calculated according to the formula:

$$h = \frac{h_{\text{opt}} \cdot \rho_{\text{ice}} \cdot \lambda}{\rho \cdot C \cdot \Delta T_{\text{opt}}},$$

- 5 where the tabulated values are: $\rho_{\text{ice}} = 0.917 \text{ kg/m}^3$ - ice density;
 $\lambda = 33 \times 10^4 \text{ J/(kg}\cdot^\circ\text{C)}$ - specific latent heat of fusion of ice;
 $\rho = 1060 \text{ kg/m}^3$ - hemangioma density; $C = 3700 \text{ J/(kg}\cdot\text{K)}$ -
heat capacity of the hemangioma; and the experimental values are:
 h_{opt} - optimum ice thickness; ΔT_{opt} - optimum difference between
10 the hemangioma temperatures before and after cooling, selected
according to the following production rule: the optimum values h_{opt}
and ΔT_{opt} equal the step values (h_i , ΔT_i) if the step coordinate
(R_i) of a point on the curve $T_i = f(R_i)$, where $\partial T / \partial R = 0$,
coincides with the hemangioma boundary (R_{gr}), where R_i is the
15 distance from the center of mass to the current point, m;
 $(\partial T / \partial R)_i$ is the derivative of the function with respect to distance;
 R_{gr} is the distance from the center of mass to the contour boundary, m;
 h_i is the i-th ice-plate thickness, m;
 ΔT_i is the i-th temperature difference before and after cooling, $^\circ\text{C}$;
20 wherein the copy material is elastic to ensure close contact and
transparent to permit marking of the hemangioma contour; the
temperature sensor is moved in increments providing at least 5
measurements, sufficient for statistical analysis; the contour of
the ice plates reproduces the hemangioma contour; their thicknesses
25 are selected from a range corresponding to the range of hemangioma
thickness values; and the number of ice plates is 5-9.
The method for determining the thickness of a flat layer is applied
to living tissue of the hemangioma type, which has a temperature
gradient relative to adjacent tissues. A transparent and elastic
30 material is placed on this thermally distinct region so that the
hemangioma contours remain visible. The contour is marked, and an
outline is cut from this marking to form a copy blank. The copy itself
is made from a denser material capable of retaining its shape when
placed on the hemangioma. The center of mass of the copy is determined,
35 and from it the direction and distance to the most distant point of
the contour are established. The temperature sensor is then fixed
above the hemangioma, and the temperature is recorded before thermal
action on the living tissue. The temperature on the hemangioma surface
is recorded stepwise along the selected direction. The unilateral
40 pulsed thermal action is produced by creating a negative heat flow
using ice plates of different thicknesses whose contact contour
reproduces the copy contour. Their thicknesses correspond to the
hemangioma thickness range, and 5-9 plates are used. After each plate
has completely melted, moisture is removed, and the temperature change
45 on the hemangioma surface is immediately recorded stepwise

along the selected direction. At least five temperature recordings are required to obtain data for statistical analysis and construction of the graph.

Figure 1 presents the graphical relationships. Symbols: T - temperature; R - distance; h_i - ice thickness; Rgr - distance from the hemangioma center of mass to the most distant point of its contour.

The technical result of the proposed method is objective identification of the surface-layer thickness in hemangioma-type living tissue with an accuracy of 5-10%. After thermal exposure and recording of the responses, the initial temperature of the measured area is restored, while its thermophysical (heat capacity), geometric (volume, area, contour, and protrusion of the hemangioma), and other parameters (surface roughness and color, including melanin and erythema pigment values) remain unchanged. In addition, the blood flow velocity before thermal exposure and after restoration of the hemangioma temperature remains unchanged, as confirmed by ultrasound data.

Thus, the proposed method for determining the thickness of a flat layer by thermal exposure of superficial hemangioma-type living tissue can be used for diagnosing such neoplasms. It avoids irradiation of the body and the associated labor costs.

(57) CLAIM

A method for determining the thickness of a flat layer, comprising subjecting the layer on one side to a pulsed heat flux that is constant over the pulse duration and uniformly distributed over the layer surface, recording the temperature change on the layer surface, and determining the layer thickness with allowance for the absorbed heat flux and the heat capacity of the layer material, characterized in that the flat layer is a layer of hemangioma-type living tissue; before the one-sided pulsed heat-flux exposure, a copying material is placed on the hemangioma, its contour is marked on the material, and a copy is made therefrom; the center of mass of the copy is determined, together with the direction and distance from the center to the most distant point of the contour; a temperature sensor is fixed above the hemangioma; the one-sided pulsed heat-flux exposure is produced by creating a negative heat flux using a set of ice plates successively placed on the hemangioma; after each plate has completely melted, moisture is removed; the temperature change on the hemangioma surface is recorded stepwise along the selected direction; and the required thickness is calculated using the formula:

$$h = (h_{\text{opt}} \cdot \rho_{\text{ice}} \cdot \lambda) / (\rho \cdot C \cdot \Delta T_{\text{opt}}),$$

where the tabulated values are:

$\rho_{\text{ice}} = 0.917 \text{ kg/m}^3$ - ice density;

$\lambda = 33 \times 10^4 \text{ J/kg}$ - specific latent heat of fusion of ice;

$\rho = 1060 \text{ kg/m}^3$ - hemangioma density; $C = 3700 \text{ J/(kg}\cdot\text{K)}$ - heat capacity of the hemangioma;

and the experimental values are: h_{opt} - optimum ice thickness; ΔT_{opt} -

5 optimum difference between the hemangioma temperatures before and after cooling, selected according to the following production rule:

the optimum values h_{opt} and ΔT_{opt} are equal to the step values (h_i , ΔT_i)

if the step coordinate R_i of a point on the curve $T_i = f(R_i)$, at which

10 the derivative $\partial T / \partial R = 0$, coincides with the hemangioma boundary R_{gr} ,

where R_i is the distance from the center of mass to the current point, m;

$(\partial T / \partial R)_i$ is the derivative of the function with respect to distance;

15

R_{gr} is the distance from the center of mass to the contour boundary, m;

h_i is the i-th ice-plate thickness, m;

20 ΔT_i is the i-th temperature difference before and after cooling, °C;

wherein the copy material is elastic to ensure close contact and transparent to permit marking of the hemangioma contour; the temperature sensor is moved in increments providing at least five measurements; the contour of the ice plates reproduces the hemangioma contour; their thicknesses are selected from a range corresponding to the hemangioma thickness range; and 5-9 ice plates are used.

30

35

40

45

METHOD FOR DETERMINING THICKNESS OF FLAT LAYER

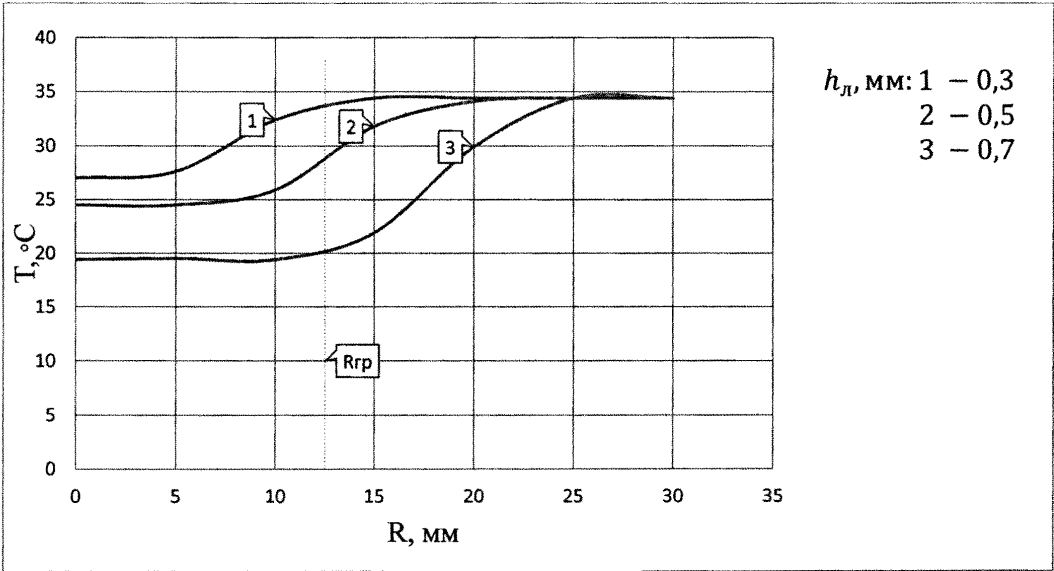


Fig. 1